A cord is used to vertically lower an initially stationary block of mass M at a constant downward acceleration of g/4. When the block has fallen a distance d, find the following:

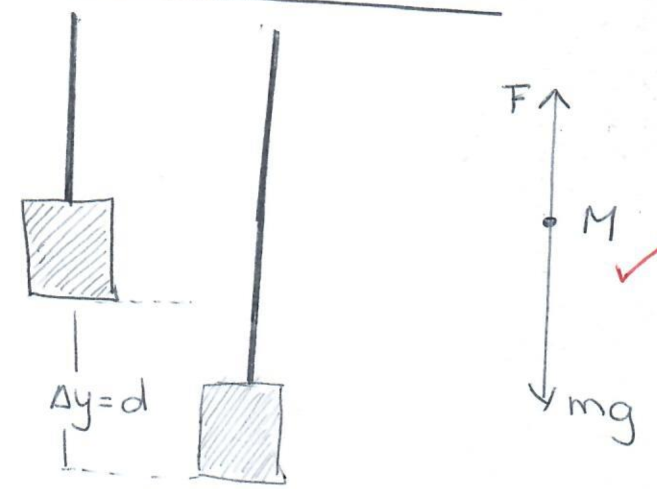
(a) the work done by the cord's force on the block,
(b) the work done by the gravitational force on the block,
(c) the kinetic energy of the block, and
(d) the speed of the block.

'n Koord word gebruik om 'n blok van massa M, wat aanvanklik in rus is, vertikaal te laat sak teen 'n afwaartse versnelling van g/4. Wanneer die blok 'n afstand d geval het, bereken die volgende:

(a) die arbeid verrig deur die koord se krag op die blok,
(b) die arbeid verrig deur die gravitasiekrag op die blok,
(c) die kinetiese energie van die blok, en
(d) die spoed van die blok.

(10)

Tutorial Test 6-1



$$\Sigma F_y = Ma = -\frac{Mg}{4} = F - Mg$$

$$\therefore F = M(g - g/4)$$

a) $W = \bar{F} \cdot \bar{d} = Fd\cos\theta$ $(\theta = 180°)$

$$= -Fd = -M(g - g/4)d$$

$$= -\frac{3}{4}Mgd$$

b) $W = \bar{F}_g \cdot \bar{d} = Mgd\cos 0°$

$$= Mgd$$

c) $K_i = 0$

$$\Delta K = W_{net} = W_F + W_g$$

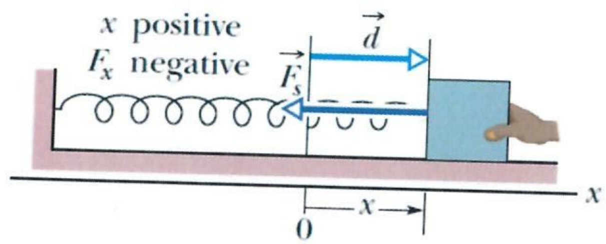
$$K_f - K_i = -\frac{3}{4}Mgd + Mgd$$

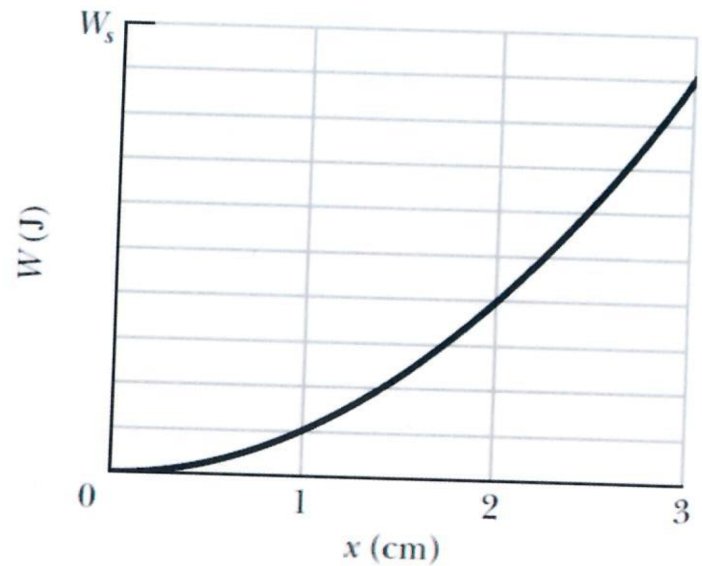
$$K_f = \frac{1}{4}Mgd$$

d) $K_f = \frac{1}{2}Mv_f^2 = \frac{1}{4}Mgd$

$$v_f = \sqrt{\frac{gd}{2}}$$

In the arrangement in the figure, we gradually pull the 3.0 kg block from $x = 0$ to $x = +3.0$ cm, where it is stationary. The graph gives the work that our external force does on the block. The scale of the graph's vertical axis is set by $W_s = 1.0$ J. We then pull the block out to $x = +5.0$ cm and release it from rest. How much work does the spring do on the block when the block moves from $x_i = +5.0$ cm to (a) $x = 2.0$ cm and (b) $x = -2.0$ cm?





In die opstelling in die figuur, trek ons die 3.0 kg blok gelydelik van $x = 0$ na $x = +3.0$ cm, waar dit stilstaan. Die grafiek gee die arbeid wat die eksterne krag verrig op die blok. Die skaal van die grafiek word gestel sodat $W_s = 1.0$ J. Die blok word dan verder getrek tot $x = +5.0$ cm en word daar gelos vanuit 'n posisie van rus. Hoeveel arbeid word verrig deur die veer op die blok wanneer die veer van $x_i = +5.0$ cm tot (a) $x = 2.0$ cm en (b) $x = -2.0$ cm beweeg?

(10)

Tutorial Test 6-2

From the graph determine k.

At $x = 3.0\,cm$, $W_S = 0.9\,J$
$x_i = 0.0\,cm$, $W_S = 0\,J$

$W_{ext} = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2$

x W_S

For $x = 3.0\,cm$ $W_s = 0.9\,J$ ✓✓ choose a point on the graph

Then $W_{ext} = \frac{1}{2}k\left(x_f \right)^2 - (0)) = 0.9\,J$

$K = 2\frac{W_{ext}}{x_f^2}$ ✓ $= \frac{2(0.9\,J)}{(3 \times 10^{-2}\,m)}$

$K = 2.0 \times 10^3\,N/m$ ✓✓

a) If $x_i = 5.0\,cm$ and $x_f = 2.0\,cm$

$W_s = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$

$W_s = \frac{1}{2}k(x_i^2 - x_f^2)$ ✓

$= \frac{1}{2}(2.0 \times 10^3\,N/m)\left[(5 \times 10^{-2}\,m)^2 - (+2.0 \times 10^{-2}\,m)^2\right]$

$= 2.1\,J$ ✓

✓ must be in m.

b) If $x_i = 5.0\,cm$ and $x_f = -2.0\,cm$

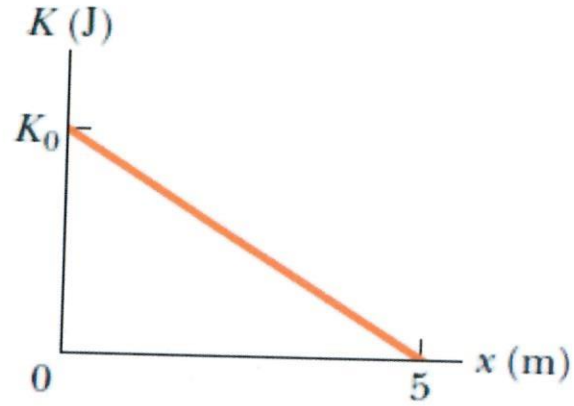
$W_s = \frac{1}{2}k(x_i^2 - x_f^2)$

$= \frac{1}{2}(2.0 \times 10^3\,N/m)\left[(5 \times 10^{-2}\,m)^2 - (-2.0 \times 10^{-2}\,m)^2\right]$ ✓

-ve

$= 2.1\,J$ ✓

An 8.0 kg object is moving in the positive direction of an x axis. When it passes through $x = 0$, a constant force directed along the x-axis begins to act on it. The figure gives its kinetic energy K versus position x as it moves from $x = 0$ to $x = 5.0$ m; $K_0 = 30.0$ J. The force continues to act. What is the change in the kinetic energy with respect to $x = 0$ m and determine v when the object moves back through $x = -3.0$ m?



'n 8.0 kg voorwerp beweeg in die rigting van die positiewe x-as. Wanneer dit deur $x = 0$ beweeg, begin 'n konstante krag op die voorwerp inwerk. Die figuur wys die kinetiese energie K teen die posisie x, soos dit van $x = 0$ tot $x = 5.0$ m beweeg; $K_0 = 30.0$ J. Die krag hou aan om in te werk op die voorwerp. Wat is die verandering in die kinetiese energie ten opsigte van $x = 0$ m en bereken die voorwerp se spoed, v, wanneer dit terug beweeg deur $x = -3.0$ m?

Tutorial Test 6-3

$K_i = 30.0\,J$ $\quad$ $K_f = 0\,J$ $\quad$ $m = 8.0\,kg$

$\Delta x = 5.0\,m$

First determine the magnitude of the Force and directio

$W = \bar{F} \cdot \Delta\bar{x} = F\Delta x \cos\theta$

$W_{ext} = \Delta K$

$\therefore F\Delta x \cos\theta = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$F\cos\theta = \dfrac{-\frac{1}{2}mv_i^2}{\Delta x} = -\dfrac{30.0\,J}{5.0\,m}$

$= -6.0\,N$

Force is in the $-x$ direction.

When object moves through $-3.0\,m$:

$K_i = 30.0\,J$, $K_f = ?$ $\quad$ $F_x = -6\,N$ $\quad$ $\Delta x = -3.0\,m$

$W = \bar{F} \cdot \Delta\bar{x} = F\Delta x \cos\theta$, $\theta = 0°$

$= (-6\,N)(-3.0\,m)$

$= 18.0\,J = \Delta K$

$W = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$v_f = \sqrt{\dfrac{2(W + K_i)}{m}}$

$= 3.46\,m/s$

<u>OR</u>.

You can take $K_i = 0$, $K_f = ?$, $F = -6\,N$,

$\Delta x = (-3 - 5)\,m = -8\,m$

$W = \Delta K = 48.0\,J$

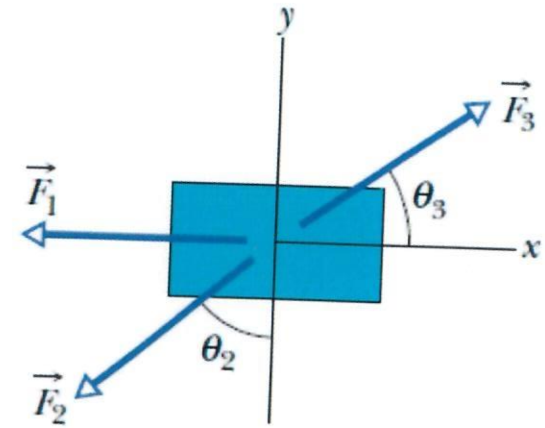
$K_f = 48.0\,J$

The figure shows an overhead view of three horizontal forces acting on a 5.00 kg cargo crate that was initially stationary but now moves across a frictionless floor. The force magnitudes are $F_1 = 3.00$ N, $F_2 = 4.00$ N, and $F_3 = 10.0$ N, and the indicated angles are $\theta_2 = 50.0°$ and $\theta_3 = 35.0°$.

(a) In what direction will the crate move?

(b) What is the net work done on the crate by the three forces during the first 4.00 m of displacement?

(c) What is the speed of the crate after it has moved 4.0 m?



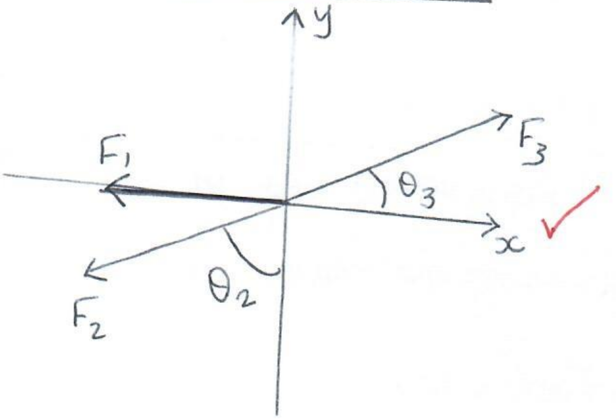
Die figuur wys die oorhoofse aansig van drie horisontale kragte wat op 'n 5.00 kg krat inwerk. Die krat was aanvanklik in rus, maar gly nou oor 'n wrywinglose vloer. Die groottes van die kragte is $F_1 = 3.00$ N, $F_2 = 4.00$ N, en $F_3 = 10.0$ N, en die hoeke wat aangedui is, is $\theta_2 = 50.0°$ en $\theta_3 = 35.0°$.

(a) In watter rigting sal die krat beweeg?

(b) Wat is die netto arbeid wat verrig is op die krat deur die drie kragte tydens die aanvanklike verplasing van 4.0 m?

(c) Wat is die spoed van die krat nadat dit 4.0 m beweeg het?

Tutorial Test 6-4



All forces are constant and therefore

$$W = F_{net}\,\Delta r$$

$$\Sigma F_x = ma_x = -F_1 - F_2 \sin\theta_2 + F_3 \cos\theta_3$$

$$\Sigma F_x = -3.00\text{ N} - 4.00\text{ N} \sin 50° + 10.0\text{ N} \cos 35°$$

$$= 2.13\text{ N}$$

$$\Sigma F_y = F_3 \sin 35° - F_2 \cos 50°$$

$$= 3.16\text{ N}$$

$$F_{net} = \sqrt{F_x^2 + F_y^2} = 3.8\text{ N}$$

$$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right) = 56°$$

above $+x$ axis

a)

The crate will move in the direction 56° above $+x$ axis

b) $W_{net} = F_{net}\,\Delta r \cos\theta \quad ; \quad \theta = 0°$

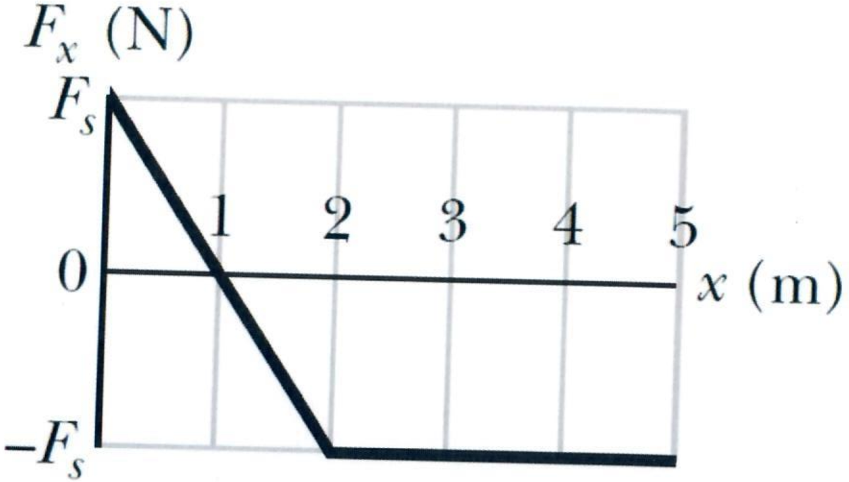
$$= 3.8\text{ N}(4.00\text{ m}) = 15.2\text{ J}$$

c) $W_{net} = \Delta K = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$$= \frac{1}{2}mv_f^2 \quad ; \quad v_i = 0\text{ m/s}$$

$$v_f = \sqrt{\frac{2W}{m}} = \sqrt{\frac{2(15.2\text{ J})}{5.00\text{ kg}}} = 2.47\text{ m/s}$$

The only force acting on a 2.0 kg body as the body moves along an x axis varies as shown in the figure. The scale of the figure's vertical axis is set by $F_s = 4.0$ N. The velocity of the body at $x = 0$ is 4.0 m/s.



(a) What is the kinetic energy of the body at $x = 3.0$ m?

(b) At what value of x will the body have a kinetic energy of 8.0 J?

(c) What is the maximum kinetic energy of the body between $x = 0$ and $x = 5.0$ m?

Die enigste krag, wat inwerk op 2.0 kg 'n voorwerp wat langs die x-as beweeg, varieer soos aangedui in die figuur. Die skaal op die figuur se vertikale as word bepaal deur F_s=4.0 N. Die snelheid van die voorwerp by $x = 0$ is 4.0 m/s.

(a) Wat is die kinetiese energie van die voorwerp by $x = 3.0$ m?

(b) By watter x waarde sal die voorwerp 'n kinetiese energie van 8.0 J hê?

(c) Wat is die maksimum kinetiese energie van die voorwerp tussen $x = 0$ en $x = 5.0$ m?

Tutorial Test 6-5

a) From graph - Work done by force between $x = 0$ and $x = 2m$ is zero: $W_{0-2} = 0J$

Area under the graph between graph & x-axis is zero.

Work done between $x = 2m$ and $x = 3.0m$ is:

$$W_{2-3} = F\Delta x = -4N(1m) = -4J.$$

$$W_{Total} = W_{0-2} + W_{2-3} = \Delta K = K_f - K_i$$

$$K_f = W_{0-2} + W_{2-3} + K_i$$

$$= 0 + (-4J) + \frac{1}{2}(2kg)(4m/s)^2$$

$$= -4J + 16J = 12J$$

b) $$W_{total} = W_{0-2} + W_{2-?} = K_f - K_i$$

$$W_{2-?} = F\Delta x = 8J - 16J$$

$$= -8J$$

$$\Delta x = \frac{-8J}{-4N}$$

$$\Delta x = 2m$$

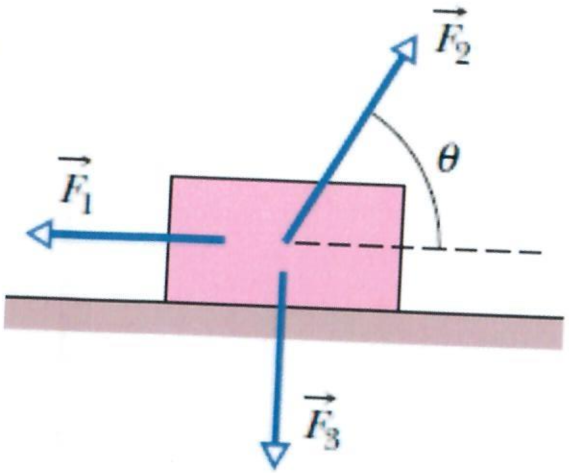
This distance is from 2m; therefore $x_f = 4m$

c) Kinetic Energy will increase when the work done is positive. This only happens from $x = 0m$ to $x = 1m$.

At $x = 1m$, $W_{0-1} = \Delta K = K_f - K_i$

$$K_f = W_{0-1} + K_i = \frac{1}{2}(1m)(4N) + 16J$$

$$= 18J$$



1. The figure on the left shows three forces applied to a crate that moves to the left by a distance 3.00 m over a frictionless floor. The force magnitudes are $F_1 = 5.00$ N, $F_2 = 9.00$ N, and $F_3 = 3.00$ N, and the indicated angle is $\theta = 60.0°$.

a) What is the net work done on the crate by the three forces during the displacement?

b) By how much does the kinetic energy of the crate change? Is the change in the **velocity** positive or negative?

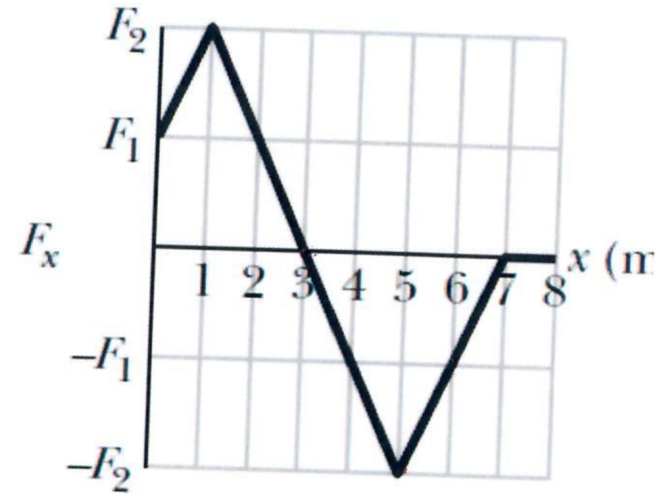
Die figuur links dui drie kragte aan wat inwerk op 'n krat wat links beweeg oor 'n afstand van 3.00 m oor 'n wrywinglose vloer. Die kragte het groottes $F_1 = 5.00$ N, $F_2 = 9.00$ N, en $F_3 = 3.00$ N, en die hoek $\theta = 60.0°$.

a) Wat is die netto arbeid verrig op die krat deur die drie kragte oor die verplasing?

*b) Met hoeveel verander die kinetiese energie van die krat? Is die verandering in die **snelheid** positief of negatief?*

2. The figure on the right gives the x-component of a force that can act on a particle. If the particle begins at rest at $x = 0$, what is its coordinate when a) it has its greatest kinetic energy, and b) when it is stationary again?

Die figuur regs dui die x-komponent van 'n krag wat inwerk op 'n deeltjie. Indien die deeltjie met rus begin by $x = 0$, wat is die koordinaat wanneer a) dit die grootste kinetiese energie het, en b) wanneer dit weer stilstaan?



Tutorial 6-6

1. $\Sigma F_x = F_{2x} - F_1 = F_2 \cos 60° - F_1$

$= 9.00\text{ N} \cos 60° - 5.00\text{ N}$

$= -0.50\text{ N}$

$\Sigma F_y = F_{2y} - F_3 = F_2 \sin 60° - F_3$

$= 9.00\text{ N} \sin 60° - 3.00\text{ N}$

$= 4.81\text{ N}$

a) $W = F_x \Delta x = -0.50\text{N}(-3.00\text{m})$

$= 1.5\text{ J}.$ must be positive

b) $W = \Delta K = 1.5\text{ J}$

$\Delta \bar{v}$ is negative

---

2. a) K_{max} at 3 m

F is positive (a is +) up to 3m.

b) $V = 0$ at 6m

Area under curve = W

Area = 0 at 6m.